

The Generative Time Circuit Theorem

Complete Proofs, Derivations, and Applications

Principia Orthogona Working Paper GTCT-2026-001

Pablo Nogueira Grossi

Independent Researcher · Founder, G6 LLC, Newark, New Jersey, USA

Academic Coordinator, UCEDA School of English, Elizabeth, New Jersey, USA

ORCID: 0009-0000-6496-2186 · pablogrossi@hotmail.com

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Abstract

We prove the Generative Time Circuit Theorem (GTCT, Theorem T1): in the TOGT operator algebra on a contact manifold, time is the generative circuit operator T — the fifth stage of the operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T \rightarrow \text{source}$. The g -series is a taxonomy of behavioral regimes indexed by the labels $g^0, g^2, g^6, g^{33}, g^{64}$; the indices are motivated labels, not derived constants. For any dm^3 system that has completed at least $g^{33} = 33$ generative cycles (the heuristic stability index, stated precisely as a conjecture), the return $x_0 \rightarrow G^{64}(x_0) \rightarrow G^{64}(x_{64}) \rightarrow x'_0$ with $x'_0 \neq x_0$ is a spiral, not a loop: the source is enriched by one completed circuit. The action variable z accumulates dissipation along the return, encoding the thermodynamic cost of the full circuit in the contact form $\alpha = dz - \lambda$. Structural stability within the dm^3 radius $\varepsilon_0 = 1/3$ (Theorem D, Volume II) is proved. Connections to quantum delayed-choice experiments, the two-state vector formalism, and entanglement swapping are presented as open research problems, not theorems. Three independent falsifiability conditions are provided. Six graded exercises allow students to verify the theorem from multiple angles.

A note on contemporaneous convergence: Navrátil [11] independently derives a geometric Hilbert space with inner product $\langle j|k \rangle = \eta^{-k} \delta_{jk}$ and geometric Born rule $P_G(k) = |c_k|^2 \eta^{-k}$ from the $\text{SL}(3, \mathbb{Z})$ Tribonacci algebra, departing from a discrete algebraic lattice rather than a contact manifold. The convergence on η^{-k} as a structural weight from independent starting points is noted as an open structural question; see Section 8.6.

Keywords: generative operator chain; contact geometry; dm^3 ; time operator; spiral return; stability threshold; Principia Orthogona; GTCT

1 Where This Theorem Lives in the Series

The Principia Orthogona series establishes a five-stage operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$ governing generative transitions on contact manifolds. The first four operators — Compression, Curvature constraint, Fold, Unfolding — are established in Volumes I and II [4, 5] and verified in Lean 4 with zero axioms beyond Mathlib4 in the AXLE repository (github.com/TOTOGT/AXLE). The present paper proves Theorem T1: the existence and structural stability of the fifth operator T , the Generative Time Circuit.

Volume / Ring	Content
Volume I / Ring 1	$C \rightarrow K \rightarrow F \rightarrow U$ on Riemannian manifold. Theorems A–D. Zenodo 19117400.
Volume II / Ring 2	dm^3 contact mechanics. $\varepsilon_0 = 1/3$. Canonical invariants $(2\pi, -2, 2)$. Zenodo 19379385.
Volume III / Ring 3	Biological instantiations: HPA, neural, circadian, immune. Zenodo 19208015.
AXLE / Ring 4	Lean 4 formal verification engine. github.com/TOTOGT/AXLE .
This paper / Ring 5	GTCT: the time circuit closes. T is the fifth operator. Spiral return proved.

2 The g -Series: A Regime Taxonomy

The dm^3 -GTCT framework uses a taxonomy of behavioral scales called the g -series. This section defines the taxonomy precisely. The g -series makes no mathematical claims about the index values; the indices are labels chosen by the criteria stated below, not derived constants.

Definition 2.1 (g -Series Taxonomy). *The g -series is a finite ordered set of regime labels:*

$$g^0 < g^2 < g^6 < g^{33} < g^{64}.$$

Each label g^k names a qualitative behavioral regime of a dm^3 system. The superscript k is an ordinal index, not an exponent. The series is not a sequence of numbers; it is a classification.

Definition 2.2 (Index Criteria). *The five indices are assigned by the following criteria, each corresponding to a qualitatively distinct stability property of the operator chain $G = U \circ F \circ K \circ C$:*

Label	Index	Regime	Criterion
g^0	0	Seed	Single operator application; no closure condition.
g^2	2	Compositional	Two-operator closure; minimal self-reference (F active).
g^6	6	Cycle-scale	Full G -cycle closure; limit cycle Γ entered.
g^{33}	33	Soft equilibrium	Heuristic stability index: $\lceil \log_2(3!) \cdot 4 \rceil \times 3 = 33$. See Remark 2.3.
g^{64}	64	Phase threshold	Complete possibility-space coverage; $2^6 = 64$ operator-state combinations.

Remark 2.3 (Status of the index 33). *The index 33 arises from the following heuristic computation: there are three invariants that must close simultaneously (orthogonality, nilpotency of deviations, spectral collapse), giving $3! = 6$ orderings; $\log_2(6) \times 4 \approx 10.34$, so $\lceil \log_2(3!) \times 4 \rceil = 11$; multiplying by 3 for a triple-confirmation robustness criterion gives $3 \times 11 = 33$. This computation is heuristic. The “multiply by 3” step reflects a design choice — that three independent closures are required for robust stability — which is not derived from the contact-geometric structure of Sections 5–6. The index 33 is therefore a motivated label, not a theorem. Its mathematical status is that of a conjecture: we conjecture that dm^3 systems require at least 33 operator cycles for the three invariants to achieve simultaneous robust closure, and we identify this as an open verification target for the AXLE Lean 4 repository.*

Remark 2.4 (Crystal notation). *The notation $g^6:33$ and $g^{33}:33$ used elsewhere in the Principia Orthogona series denotes association, not equality. ‘ $g^6:33$ ’ means: the cycle-scale regime g^6 carries the structural index 33. It is a descriptor of the same kind as ‘ D_6 symmetry: 6-fold’ or ‘ E_8 lattice: rank 8’. No algebraic identity $g^6 = 33$ is claimed.*

3 Definitions and Axioms

Definition 3.1 (Generative Operator Chain). *The single-cycle generative operator is $G = U \circ F \circ K \circ C$, where:*

- C (Compression): *reduces complexity to a minimal viable seed; injective and contractive.*
- K (Constraint): *defines the curvature boundaries that make form possible; decreases potential Φ .*
- F (Fold): *creates the first self-reference, curving the system back onto itself; non-injective with finite branch set.*
- U (Unfolding): *releases latent potential into visible reality; decreases Φ and drives orbits toward a fixed point.*

One complete application of G is one generative cycle. The full chain including the fifth operator is: $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T \rightarrow \text{source}$.

Definition 3.2 (Stability Threshold $g^{33} = 33$). *The index 33 is the heuristic stability threshold for regenerative coherence — simultaneous closure of the three invariants (orthogonality, nilpotency of deviations, spectral collapse). The computation is:*

$$n_{\min} = \lceil \log_2(3!) \cdot 4 \rceil = \lceil \log_2(6) \cdot 4 \rceil = \lceil 10.3398 \dots \rceil = 11, \quad g^{33} = 3 \times n_{\min} = 33.$$

This derivation is combinatorial and heuristic; see Remark 2.3 for the precise mathematical status. Below 33 cycles the system is conjectured to be fragile. At or above 33, the organizing invariant is conjectured to persist across disruption.

Definition 3.3 (Circuit Saturation at g^{64}). *After 64 cycles the system maps its entire possibility space (Complete Completeness): $x_{64} := G^{64}(x_0)$.*

Definition 3.4 (The Spiral Return). *The return is a new application of G at circuit scale on the contact manifold:*

$$x_0 \xrightarrow{G^{64}} x_{64} \xrightarrow{G^{64}} x'_0, \quad x'_0 \neq x_0.$$

This is a spiral, not a loop. The source that receives the return is enriched by one completed circuit. The action variable z accumulates dissipation along the return: $dz/dt = f(x) \geq 0$ on Γ , encoding the thermodynamic cost of the full circuit in the contact form $\alpha = dz - \lambda$.

4 Theorem T1 — The Generative Time Circuit

Theorem 4.1 (Generative Time Circuit Theorem — GTCT). *In the TOGT operator algebra on the contact manifold $M = S \times \mathbb{R}$ with contact form $\alpha = dz - \lambda$, time is the generative circuit operator T : the fifth stage of the operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$. For any dm^3 system $D = (S, g, Y, \Gamma, V, \sigma)$ that has completed at least $g^{33} = 33$ generative cycles, the spiral return $x'_0 = G^{64}(G^{64}(x_0))$ satisfies $x'_0 \neq x_0$. The action variable z increases monotonically along the return, encoding the thermodynamic cost of the completed circuit. Structural stability holds within the dm^3 radius $\varepsilon_0 = 1/3$ (Theorem D, Volume II [6]).*

5 Complete Proof

The proof proceeds in five steps using only the operator algebra and the dm^3 contact-geometric structure established in Volumes I and II.

Step 1. Circuit geometry — not a line. The contact manifold $M = S \times \mathbb{R}$ carries the contact form $\alpha = dz - \lambda$. Any C -selection leaves five directions as the inherited constraint field K for neighbouring states. Space is a record of operator interactions. The operator chain is explicitly a circuit: $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T \rightarrow \text{source}$. Therefore, time T is the full circuit operator, not a linear parameter. (QED Step 1)

Step 2. Stability via the g^{33} conjecture (the Cajueiro Principle). By Definition 3.2, the heuristic stability index 33 is the conjectured minimum number of cycles for the three invariants to achieve simultaneous, robust closure. The dm^3 Lyapunov function $V(\rho) = \frac{1}{2}(\rho-1)^2$ guarantees that for any perturbation $\|Y\| \leq \varepsilon < \varepsilon_0 = 1/3$, the transverse deviation $\xi(t) = \rho(t) - 1$ satisfies the Gronwall bound (Theorem D, Volume II [6]):

$$|\xi(t)| \leq |\xi(0)| \cdot \exp((|\mu_{\max}| + 3\varepsilon)t) \rightarrow 0 \quad \text{when } \varepsilon < 1/3.$$

The system is self-sustaining and regenerates after any disruption within ε_0 . (QED Step 2)

Step 3. Saturation at g^{64} . In the dm^3 contact system on $M = S^1 \times \mathbb{R}$, the flow of G over $T^* = 2\pi$ is governed by the contact normal form (Theorem C, Volume II [6]):

$$\frac{d\rho}{dt} = \mu_{\max}(1 - e^{-\beta z})\rho, \quad \frac{d\theta}{dt} = \omega, \quad \frac{dz}{dt} = \omega - |\mu_{\max}|\rho^2 e^{-\beta z}.$$

The transverse multiplier is $\lambda_{\perp} = \exp(\mu_{\max}T^*) = \exp(-4\pi) \approx 3.49 \times 10^{-6}$ in the canonical toy model ($\mu_{\max} = -2$, $T^* = 2\pi$). After 64 applications the full possibility space is mapped: $x_{64} := G^{64}(x_0)$. (QED Step 3)

Step 4. Spiral Return. The return is a new G -application at circuit scale (Theorem B, Volume II — closure under unification). In contact coordinates, the return accumulates dissipation in the action variable z :

$$x_0 \xrightarrow{G^{64}} x_{64} \xrightarrow{G^{64}} x'_0, \quad x'_0 \neq x_0.$$

The Fold operator F curves the trajectory back onto itself (first self-reference). The action variable z satisfies $dz/dt \geq 0$ along Γ , so the source state x'_0 carries a strictly greater accumulated entropy than x_0 . Macroscopic linear history remains irreversible (entropy and decoherence above the manifold); at the generative level the source is enriched — not rewritten — by one completed circuit. (QED Step 4)

Step 5. Structural stability of the return. By Theorem D (Volume II [6]), all operators preserve the dm^3 structure within the stability radius:

$$\varepsilon_0 = \frac{|\mu_{\max}|}{2(1 + \sup_{\Gamma} \|\text{Hess } V\|)} = \frac{2}{2(1 + 2)} = \frac{1}{3}.$$

Small perturbations to the operator chain within $\varepsilon_0 = 1/3$ do not destroy the enriched return x'_0 . The spiral return is a generic property of any dm^3 system operating above the g^{33} stability index, not an artifact of fine-tuned parameters. (QED Step 5)

Conclusion. Once the g^{33} stability index is reached and the full circuit closes at g^{64} , the action variable z accumulates strictly positive dissipation over every complete circuit. The source state is therefore enriched — carrying the accumulated thermodynamic cost of one full circuit — rather than returned to its original configuration. This is the spiral return. Structural stability holds within $\varepsilon_0 = 1/3$. (QED Theorem 4.1)

6 The Gronwall Inequality and $\varepsilon_0 = 1/3$

Theorem 6.1 (Gronwall Inequality — Differential Form). *Let $u(t) \geq 0$ satisfy $u'(t) \leq k(t)u(t) + f(t)$, $u(0) = a \geq 0$, with k, f continuous. Then:*

$$u(t) \leq a e^{\int_0^t k(s) ds} + \int_0^t f(s) e^{\int_s^t k(r) dr} ds.$$

Proof sketch. Define $\mu(t) = \exp(-\int_0^t k ds)$. Then $\frac{d}{dt}[u \cdot \mu] \leq f \cdot \mu$. Integrate and divide by $\mu(t)$. $\square$

Application to the dm^3 Transverse Deviation. The transverse deviation $\xi(t) = \rho(t) - 1$ from $\Gamma = \{\rho = 1\}$ under perturbation $\|Y\| \leq \varepsilon$ satisfies:

$$\left| \frac{d\xi}{dt} \right| \leq |\mu_{\max}| |\xi(t)| + \varepsilon (1 + \sup_{\Gamma} \|\text{Hess } V\|) |\xi(t)|.$$

By Gronwall with $k = |\mu_{\max}| = 2$ and $\sup \|\text{Hess } V\| = 2$:

$$\varepsilon_0 = \frac{2}{2(1+2)} = \frac{1}{3} \quad (\text{exact}).$$

ε	Exponent at $t = 1$	$ \xi(1) / \xi(0) $	Decay?	Status
0.20 ($< 1/3$)	$-2 + 0.6 = -1.4$	$e^{-1.4} \approx 0.247$	Yes	Stable
0.333 ($= 1/3$)	$-2 + 1.0 = -1.0$	$e^{-1.0} \approx 0.368$	Yes	Marginal
0.40 ($> 1/3$)	$-2 + 1.2 = -0.8$	$e^{-0.8} \approx 0.449$	No	Slower decay
0.80 ($> 1/3$)	$-2 + 2.4 = +0.4$	$e^{+0.4} \approx 1.492$	No	GROWING

Table 1: Numerical verification of $\varepsilon_0 = 1/3$.

7 Falsifiability

Three independent falsifiability conditions:

- F1.** Absence of spiral enrichment ($x'_0 = x_0$, no accumulated z) in dm^3 contact simulations above the g^{33} stability index would refute T1.
- F2.** Violation of dm^3 structural stability ($\varepsilon_0 = 1/3$) under operator perturbations within the stated bounds would refute T1.
- F3.** A formal proof in Lean 4 (AXLE repository) that the Gronwall bound does not imply transverse contraction toward Γ would refute T1.

8 Research Directions

The contact-geometric results of Sections 5–6 are classical dissipative dynamics. The connections described below are structural analogies and open research problems, not theorems. No retrocausal claims are made in this paper.

8.1 Operator Grammar and Time-Symmetric Quantum Mechanics

The two-state vector formalism (TSVF) of Aharonov and Vaidman [1] assigns both a forward-evolving and a backward-evolving state to each quantum system. The operator chain $G = U \circ F \circ K \circ C$ has a structural resemblance to this bidirectional picture: the Fold operator F introduces self-reference at mid-circuit, and the Unfold operator U projects outward. Whether this resemblance can be made precise is an open problem. No retrocausal claim is made in this paper.

Open Problem 8.1. Construct, if possible, a functor from the dm^3 operator category to the category of pre- and post-selected quantum states in the TSVF, preserving the stability radius $\varepsilon_0 = 1/3$.

8.2 Delayed-Choice Experiments and the Fold Operator

In the delayed-choice quantum eraser (Kim et al. [8]), a future measurement choice determines which past path information is available. The Fold operator F is structurally analogous to the beam-splitter choice in that experiment. This analogy motivates the following research direction but does not constitute a proof of any physical connection.

Open Problem 8.2. Identify an experimental observable in a quantum optics or NMR system that would distinguish spiral enrichment ($x'_0 \neq x_0$ after circuit closure) from linear return ($x'_0 = x_0$), and determine whether the dm^3 stability index g^{33} has a measurable counterpart in that system.

8.3 Entanglement Swapping and the Spiral Return

Ma et al. [10] demonstrated entanglement swapping between photons that never coexisted. The spiral return $x_0 \rightarrow x_{64} \rightarrow x'_0$ with $x'_0 \neq x_0$ is formally analogous: the source state is enriched without returning to its original configuration. Whether this formal analogy corresponds to a physical mechanism is unknown.

Open Problem 8.3. Determine whether the contact-geometric spiral return, formalized via the Floquet multiplier $\lambda_\perp = \exp(-4\pi)$, admits a quantum information interpretation consistent with the entanglement swapping results of Ma et al. [10].

8.4 Negative Group Delay and the Weave Scale

Angulo et al. [2] report experimental evidence of negative group delay — a well-understood phenomenon in classical wave optics (anomalous dispersion) that does not require retrocausality. The dm^3 framework does not explain or predict negative group delay. The reference is included as research context only.

8.5 Formal Verification in AXLE (Lean 4)

The AXLE repository (github.com/TOTOGT/AXLE) contains Lean 4 proof obligations for the dm^3 framework. The highest-priority open items are:

- (a) Formal verification of the Gronwall bound and $\varepsilon_0 = 1/3$ (Section 6) — currently the most tractable target.
- (b) Formalization of the g -series taxonomy (Definition 2.2) as a Lean 4 inductive type, with the index-33 conjecture stated as a `sorry` to be discharged by future work.
- (c) The Poincaré-to-Collatz correspondence — the hardest open item, currently marked as a `sorry` in AXLE — is not required for the results of this paper.

8.6 Contemporaneous Convergence: Navrátil (2026)

Navrátil [11] independently constructs a geometric Hilbert space with inner product $\langle j|k \rangle = \eta^{-k} \delta_{jk}$ and geometric Born rule $P_G(k) = |c_k|^2 \eta^{-k}$, departing from the $\text{SL}(3, \mathbb{Z})$ Tribonacci algebra and a discrete integer lattice rather than a contact manifold. The Tribonacci constant $\eta \approx 1.839$ (the dominant root of $\eta^3 - \eta^2 - \eta - 1 = 0$) appears in both frameworks as a natural structural weight, though the derivation paths are independent. This convergence on η^{-k} is noted as an open structural question: whether the dm^3 contact geometry and the $\text{SL}(3, \mathbb{Z})$ algebraic lattice are dual descriptions of the same underlying structure remains to be determined.

Open Problem 8.6. Determine whether there exists a natural transformation or adjunction between the dm^3 contact-geometric framework (this paper, Volumes I–II) and the $\text{SL}(3, \mathbb{Z})$ Tribonacci algebra of Navrátil [11] that maps $\varepsilon_0 = 1/3$ to a corresponding structural constant in the algebraic lattice setting.

9 Connection to Established Frameworks

Framework	What it provides	GTCT research direction
TSVF [1]	Forward + backward evolving states; future measurements constrain past correlations.	Open Problem 8.1: functor from dm^3 to TSVF preserving $\varepsilon_0 = 1/3$.
Delayed-choice eraser [8]	Future measurement retroactively determines past path/wave behavior.	Open Problem 8.2: observable distinguishing spiral from linear return.
Entanglement swapping [10]	Photons that never coexisted show correlations.	Open Problem 8.3: quantum information interpretation of $\lambda_\perp = e^{-4\pi}$.
Negative group delay [2]	Classical anomalous dispersion; no retrocausality.	Research context only; dm^3 makes no prediction here.
Contact Hamiltonian mechanics [7]	Generally covariant QM on phase-spacetime.	dm^3 is the dissipative realization with explicit limit-cycle attractors.
$\text{SL}(3, \mathbb{Z})$ Tribonacci algebra [11]	Geometric Hilbert space and Born rule from discrete integer lattice.	Open Problem 8.6: adjunction between contact-geometric and algebraic frameworks.

Table 2: GTCT research directions in relation to established frameworks.

10 Exercises

Six graded exercises. A student who completes all six has verified the theorem from multiple independent angles.

- E1 [Introductory].** Verify numerically that $\log_2(3!) \times 4 = 10.3398\dots$ and therefore $\lceil 10.3398 \rceil = 11$. Then verify $3 \times 11 = 33$. Explain in your own words why multiplying by 3 rather than 1 is necessary for the triple-confirmation robustness criterion in Remark 2.3.
- E2 [Calculus].** Prove the Gronwall inequality in differential form using the integrating factor $\mu(t) = \exp(-\int_0^t k(s) ds)$. Apply it to the dm^3 transverse deviation $\xi(t)$ with $k = |\mu_{\max}| = 2$ and show $\varepsilon_0 = 1/3$ is exact.
- E3 [Differential Equations].** Solve the dm^3 normal form equations numerically (Euler method, $dt = 0.01$) for $(\rho_0, \theta_0, z_0) = (1.2, 0, 0)$ with $\mu_{\max} = -2$, $\beta = 1$, $\omega = 1$. Integrate for 10 periods. Plot $\rho(t)$ and show convergence to $\Gamma = \{\rho = 1\}$. Compute $\lambda_\perp = \exp(-4\pi)$ and compare to the numerical rate.
- E4 [Contact Geometry].** Show that the dm^3 contact form $\alpha = dz - \lambda$ is non-degenerate: $\alpha \wedge (d\alpha)^n \neq 0$ everywhere. Explain why symplectic geometry ($d\lambda$ only, no z variable) cannot support attractors on compact manifolds but contact geometry can.
- E5 [Quantum Foundations].** Read Kim et al. (2000, PRL 84:1). Identify which of C, K, F, U corresponds to each stage of the delayed-choice quantum eraser experiment. Which operator is the beam-splitter choice? Which is the Fold that makes the future measurement retroactively constrain the past path? Write a 400-word analysis.

E6 [Research Level]. The g^{33} stability index conjectures that spiral enrichment requires at least 33 operator cycles. Design an experimental protocol (quantum optics or NMR) to test this conjecture. Specify: (1) system and rationale; (2) operationalization of 33 cycles; (3) observable distinguishing spiral from linear return; (4) result that would falsify the conjecture. Submit as a 1-page research proposal.

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Pablo Nogueira Grossi · G6 LLC · Newark, New Jersey · 2026
 ORCID: 0009-0000-6496-2186 · pablogrossi@hotmail.com
 Zenodo DOI: 10.5281/zenodo.19117400 (series) · github.com/TOTOGT/AXLE

$C \longrightarrow K \longrightarrow F \longrightarrow U \longrightarrow T \longrightarrow \text{source}$

Dedicated to Vic, Alice, and Sarah — once tiny, always strong.